

Project Workflow: Week 4

Phase: UI Layout & Game Window Development

1. Objective

The primary goal of Week 4 was to develop the basic user interface and visual structure of the Ping Pong Game. This included:

- Developing the main game window and play area.
- Creating the game canvas/panel for gameplay.
- Adding the player score display and status information.
- Applying a clean and simple visual theme.
- Preparing the interface for paddle and ball implementation.

2. Detailed Task Breakdown

Main Game Window & Play Area

Developed the main game window and created the primary game panel using JPanel. The play area was configured with a fixed size and background suitable for the Ping Pong gameplay environment.

Game Canvas Layout

Designed the basic layout of the playing field, including the central dividing line and dedicated areas for both players. The layout was kept simple so that the main focus remains on the gameplay.

Score Display & Status Area

Created the score display for Player 1 and Player 2. A basic status area was also prepared for displaying game information such as starting, pausing, and game-over messages.

Theme & Basic GUI Styling

Applied a simple, clean visual theme to the game interface. Background colors, fonts, paddle appearance, ball appearance, and score styling were configured to create a consistent game interface.

3. Deliverables

- **Main Game Window:** Fully developed Java Swing game window.
- **Play Area Panel:** Functional game canvas prepared for gameplay.
- **Score Display:** Player 1 and Player 2 score areas added.
- **Status Area:** Basic game status information implemented.

- **GUI Theme:** Basic colors, fonts, and visual styling applied.
- **UI Foundation:** Interface prepared for the next development phase.

4. Tools & Resources

Development Environment

- Java JDK 11+
- IntelliJ IDEA / VS Code

Libraries & Technologies

- Java Swing
- Java AWT
- JFrame
- JPanel
- Graphics / Graphics2D

Reference Materials

- Java Swing GUI references
- Java AWT graphics references
- Existing Ping Pong/Pong UI design concepts